



Using Bandicoot's control unit

robotic unit with an arm and four legs that enters the manhole and handles the cleaning operation. The other is a control panel unit that stays outside the manhole with the person controlling or monitoring Bandicoot.

The robotic unit has a fixed, water-proof, night-vision camera that transmits 4K resolution videos and images in real time, even in the presence of water. Various sensors help Bandicoot measure different environmental parameters like manhole dimensions, material and chemicals inside the manhole, humidity, temperature and so on. The robotic arm has five degrees of freedom with 360-degree movement capacity. A bucket unit of 18 litres capacity is attached to collect waste.

The cleaning process can be done in two modes: automatic or semi-automatic. Automatic cleaning can be applied to standard manholes. It is powered by artificial intelligence (AI) programs based on data collected by sensors. The robotic unit moves inside the manhole by itself and cleans on its own.

For non-standard manholes, the semi-automatic control is brought into play. The human being controlling the robot can use the control panel that has a display screen to position Bandicoot inside the manhole. Cleaning is done automatically.

The robot is made of resilient carbon nano-fibre. To reduce the risks of harmful gases and fluids inside the manhole, the robot has only a few electronics, is water-proof and uses pneumatics.

Learning complexity of Bandicoot's control is low. George says, "Even a person unfamiliar with

technology can use it with only a week's training. Through simple graphical user interface and local language integration, we have ensured that Bandicoot can be used by anyone." Since Bandicoot will replace humans, who were initially employed for the work, its usage is

made easy enough so that the same people can be hired to control it instead, in turn, preserving their jobs.

With the help of the Kerala Government, GenRobotics received a 5,000 sq.ft. space for production from Kerala Infrastructure Development Authority (KINFRA) Industrial Park. The team installed necessary machineries like CNC, welding machine and so on, and created a production unit that has a generation capacity of 20 units.

For the new entrepreneurs who quit their respective jobs to focus on GenRobotics, lack of funds was a major challenge initially. Going forward, Kerala Chartered Mission granted them an aid of ₹ 1 million to set up their facility for the project and create a prototype. For the prototype, they reused materials and scraps.

Procuring good quality material was also challenging. Getting the grant for the production space was time consuming too.

After deployment, the major challenge lies in creating awareness about Bandicoot.

The robot has been launched in Kerala, Tamil Nadu and recently in Andhra Pradesh. Genrobotics has also acquired an Indian patent for Bandicoot. In October, they are set to reveal the upgraded Bandicoot 2.0 in New Delhi in the presence of Prime Minister Narendra Modi under Swachh Bharat Mission.

New features in Bandicoot 2.0 will include on-screen training application on the control panel for users, increased bucket size of 25 litres, better components, improved AI and voice-driven control, which is being worked upon currently.

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